you define what attribute will determine word size and what size you want the word cloud to lay out for.

Unlike most other layouts, `cloud()` fires a custom event `"end"` that indicates it's done calculating the most efficient use of space to generate the word cloud. The layout then passes to this event the processed dataset with the position, rotation, and size of the words. We can then run the cloud layout without ever referring to it again, and we don't even need to assign it to a variable, as we do in the following listing. If we plan to reuse the cloud layout and adjust the settings, we assign it to a variable like with any other layout.

Listing 5.10 Creating a word cloud with d3.layout.cloud

```
var wordScale=d3.scale.linear().domain([0,75]).range([10,160]);

d3.layout.cloud()
  .size([500, 500])
  .words(data)
  .fontSize(function(d) { return wordScale(d.frequency); })
  .on("end", draw)
  .start();

function draw(words) {

  var wordG = d3.select("svg").append("g")
      .attr("id", "wordCloudG")
      .attr("transform","translate(250,250)");

  wordG.selectAll("text")
    .data(words)
    .enter()
    .append("text")
    .style("font-size", function(d) { return d.size + "px"; })
    .style("opacity", .75)
    .attr("text-anchor", "middle")
    .attr("transform", function(d) {
       return "translate(" + [d.x, d.y] + ")rotate(" + d.rotate + ")";
    })
    .text(function(d) { return d.text; });
};
```

Uses a scale rather than raw values for the font

Assigns data to the cloud layout using .words()

Sets the size of each word using our scale

The cloud layout needs to be initialized; when it's done it fires "end" and runs whatever function "end" is associated with.

We've assigned draw() to "end", which automatically passes the processed dataset as the words variable.

Translation and rotation are calculated by the cloud layout.

This code creates an SVG `<text>` element that's rotated and placed according to the code. None of our words are rotated, so we get the staid word cloud shown in figure 5.28.

It's simple enough to define rotation, and we only need to set some rotation value in the cloud layout's `.rotate()` function:

```
randomRotate=d3.scale.linear().domain([0,1]).range([-20,20]);

 d3.layout.cloud()
  .size([500, 500])
  .words(data)
  .rotate(function() {return randomRotate(Math.random())} )
```

This scale takes a random number between 0 and 1 and returns an angle between -20 degrees and 20 degrees.

Sets the rotation for each word